PULKIT GAUTAM

COMPUTER VISION ENGINEER

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GitHub | Portfolio | LinkedIn

SUMMARY

Computer Vision Engineer with 2+ years of experience in electro-optical systems at DRDO, specializing in MWIR thermal imaging, real-time video processing, and GPU-accelerated AI deployment. Skilled in fixed-pattern noise removal, multisensor fusion, super-resolution, image stabilization and deep learning-based target detection. Hands-on experience deploying inference pipelines on edge platforms (NVIDIA Jetson) for defense applications, with a strong foundation in FPGA-based image processing and real-time system design.

TECHNICAL SKILLS

Computer Vision & AI: Deep Learning, Object Detection, Image Classification, Image Stabilization, Multisensor Fusion, Image Super Resolution, Infrared Image Processing, Non-Uniformity Correction (NUC), Bad Pixel Replacement (BPR)
Frameworks: TensorFlow, PyTorch, OpenCV, TensorRT
Programming: Python, MATLAB, Verilog
Hardware Acceleration: NVIDIA Jetson (GPU), FPGA, CUDA
Tools: Git, ONNX, LabelImg, VoTT
Domains: Electro-Optical Systems, Thermal Imaging, Real-Time Systems, Edge AI, Defense Applications

WORK EXPERIENCE

Project Engineer

IRDE, DRDO, Dehradun | September 2024 – Present

Working in Thermal Imaging Laboratory on **Electro-Optical System Design, Integration and Testing**.

- Developed real-time computer vision pipelines for MWIR **thermal imaging systems**, encompassing AI-based target detection (SSD, YOLO, R-CNN), multisensor image fusion, and digital image stabilization for defense applications.
- Designed and deployed **GPU-accelerated** deep learning inference pipelines (Python, TensorRT) on NVIDIA Jetson edge platforms, achieving real-time processing performance.
- Implemented thermal image processing and enhancement algorithms, including Non-Uniformity Correction **(NUC)** and Bad Pixel Replacement **(BPR)**.
- Built and optimized **FPGA-based image processing** modules (Verilog) for real-time thermal video pipelines, for real-time thermal imaging systems
- Integrated multisensor payloads — thermal, visible, LRF, and GPS — and conducted **system-level calibration**, validation, and testing of electro-optical systems under operational conditions.
- Developed sensor interfacing logic, communication protocols, and system interface documentation (**EICD**) for embedded electro-optical systems.

Intern

IRDE, DRDO, Dehradun | January 2024 – June 2024

Internship at Thermal Imaging Laboratory on "***Real-Time Target Detection in Thermal Imaging Systems on GPU based Platform***"

- Built and deployed a real-time thermal target detection pipeline on NVIDIA Jetson AGX Orin.
- Created and annotated **thermal imaging datasets** for model training, validation, and evaluation.
- Implemented and benchmarked deep learning models — **YOLOv7, Faster R-CNN** using TensorFlow and PyTorch, analyzing latency vs. accuracy trade-offs for real-time thermal target detection.

PROJECTS

Deep-Learning Based Real-Time Target Detection in Thermal Imaging Systems

Implemented and benchmarked deep learning models — YOLOv7, R-CNN and Faster R-CNN using TensorFlow and PyTorch real-time thermal detection on GPU based platform.

Design and Simulation of Approximate Parallel Prefix Adder with Carry-Skip in Vivado

Designed an approximate parallel prefix adder in Verilog and analyzed accuracy vs. delay vs. hardware utilization trade-offs using Vivado simulation tools.

Object Detection and Counting by Shape and Colour using Blob Analysis in MATLAB

Implemented an object detection and counting pipeline using Blob Analysis, image segmentation, and contour detection in MATLAB.

EDUCATION

B. Tech. in Electronics and Communication Engineering **CGPA 7.54/10**

National Institute of Technology Patna | 2020 – 2024

PUBLICATIONS

- **"Hardware Implementation of Fixed-Pattern Noise Removal Algorithm for an HD Infrared Detector"**
  International Conference on Electro-Optics and Photonics (EOP), Springer Nature, 2025.

CERTIFICATIONS

The Joy of Computing using Python **2023**

Course by Prof. Sudarshan Iyengar, IIT Ropar – NPTEL

An Introduction to Artificial Intelligence **2022**

Course by Prof. Mausam, IIT Delhi – NPTEL

HTML, CSS, and JavaScript for Web Developers **2020**

Course by Johns Hopkins University – Coursera